

Introduction to differential equations

- what is a differential equation? it's an equation containing a **unknown function** and **its derivative**

ex:

unknown function: $f(x) = ?$

its derivative: $f'(x), f''(x), \dots$

}

$$f''(x) + 2f'(x) = 3f(x)$$

notes and terminology

the derivative $\frac{dy}{dx}$ of a function. $y = f(x)$ is itself another $f'(x)$ the equation we made up lets say $\frac{dy}{dx} = 0.5xy$ is called a differential equation (D.E)

an equation containing the derivative of one or more unknown functions (or dependent variable) with respect (w/r) to one or more independent variables **Classify by type, order, or linearity**

type:

1) if a D.E contains only ordinary derivatives of one or more unknown functions w/r to a single independent variable **when you are taking the derivative with respect to one variable you are in ODE territory**

ex:

$$\frac{dy}{dx} = x^2 - y^2$$

2) when you begin involving partial derivatives ($\frac{\partial y}{\partial x}$) of one or more unknown function of two or more independent variables **when your function is a partial derivative you are in PDE territory**

ex:

$$\frac{\partial y}{\partial x} = x^2 + y^2$$

examples:

$$\frac{dx}{dt} + \frac{dy}{dt} = 5x + 3y \rightarrow \text{this is an ODE}$$

an ODE can contain more than one unknown function

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 5 \rightarrow \text{this is a PDE}$$

Notice how you have partial derivatives here

classification by ORDER

ORDER of an DE (ODE or PDE) is the order of the highest derivative in the equation

example:

$$\frac{d^2 y}{dx^2} + 8 \left(\frac{dy}{dx} \right)^5 - 4y = e^{5x}$$

↑ ↑
2nd-order first order

first-order ODE can sometimes look like this:

$$M(x,y) dx + N(x,y) dy = 0$$

Classification by linearity

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

↓
1st order

$$a_2(x) \frac{d^2y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

↓
2nd-order

linear ODE

- 1) the dependent variable y and all its derivatives $y', y'', \dots, y^{(n)}$ are the 1st degree, power of each term, y is 1
- 2) the coefficient $a_0, a_1, \dots, a_n$ of $y, y', \dots, y^{(n)}$ depends at most on the independent variable x

$$a_n(x) \frac{d^n y}{dx^n} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

Simpler terms

- 1) y and all of its derivatives can only have **pure functions of x** in front of y and its derivatives
- 2) y and all of its derivatives are all to the first power

a nonlinear ODE is one that is not linear usually dependent and derivatives are $\cos(y)$ or e^y , $\tan(y)$

examples:

$$(y+x) dx + 7x dy = e^x$$

this DE is non-linear because on the first term it depends on " y " we said y and its derivatives can only have pure functions of " x " in front of them

Intuition

- Linear = everything involving y is added or multiplied by functions of x only
- Nonlinear = y interacts with itself or appears in a nonlinear way

Quick Tricks to Spot Nonlinearity

- Powers of $y > 1$ → Nonlinear
- Products of y and derivatives → Nonlinear
- y inside a function ($\sin(y)$, e^y , $\ln(y)$) → Nonlinear
- Derivatives multiplied together ($y' \cdot y'$) → Nonlinear

$$\frac{d^2y}{dx^2} + \cos(y) = 0$$

this has a $\cos(y)$ so it cannot be a linear DE so its non-linear

$$y'' - 5y + y = 0$$

this here has pure functions of "x" in front all of the y's here 5 is a constant its a special case of a function of x also y and its derivatives are to the first power even y'' that just a second derivative so this is a 2nd-order linear function

Solution of an ODE

any function $f(x)$, defined on a interval "I", has at least n amount derivatives, which can be substituted into ODE, implies a solution exists of the equation

note: a solution of a DE that is identically zero on the interval "I" is said to be a trivial solution

examples

Verify that the function given is an explicit solution of the DE

$$\frac{dy}{dt} + 20y = 24$$

$$y = \frac{6}{5} - \frac{6}{5}e^{-20t}$$

all we are doing over here is taking the derivative of the given y and plugging it in to try to verify if the given solution is correct we will know if its correct if both sides are equal

$$y' = -\frac{6}{5}(-20)e^{-20t} \Rightarrow 24e^{-20t}$$

and now just plug in y' and y to see if you have a solution

$$\frac{dy}{dt} + 20y \Rightarrow (24e^{-20t}) + 20\left(\frac{6}{5} - \frac{6}{5}e^{-20t}\right)$$

$$\frac{dy}{dt} + 20y \Rightarrow \cancel{24e^{-20t}} + 24 - \cancel{24e^{-20t}}$$

so the left side is equal to 24 and the right side is also equal to 24

$$\text{left side} = \text{right side}$$

$$24 = 24$$

so, $y = \frac{6}{5} - \frac{6}{5}e^{-20t}$ is a valid solution for the D.E

example #2

$$\overbrace{(y-x)y'}^{\text{left}} = \overbrace{y-x+8}^{\text{right}}, \quad y = x + 4\sqrt{x+2}$$

$$\begin{aligned} y' &= 1 + 4(x+2)^{-1/2} \\ &= 1 + \frac{4}{2}(x+2)^{-1/2} \\ &= 1 + \frac{2}{\sqrt{x+2}} \end{aligned}$$

Plug into the equation:

$$(y-x)y' \Rightarrow (\cancel{x} + 4\sqrt{x+2} - \cancel{x}) \left(1 + \frac{2}{\sqrt{x+2}}\right)$$

$$(y-x)y' \Rightarrow (4\sqrt{x+2}) \left(1 + \frac{2}{\sqrt{x+2}}\right)$$

$$(y-x)y' \Rightarrow 4\sqrt{x+2} + \frac{8\sqrt{x+2}}{\sqrt{x+2}}$$

$$(y-x)y' \Rightarrow 4\sqrt{x+2} + 8$$

left side

$$\begin{array}{lcl}
 y - x + 8 & \Rightarrow & (\cancel{x} + 4\sqrt{x+2}) - \cancel{x} + 8 \\
 y - x + 8 & \Rightarrow & 4\sqrt{x+2} + 8
 \end{array}
 \quad \left. \vphantom{\begin{array}{l} \\ \end{array}} \right\} \text{right side}$$

left side = right side

$$4\sqrt{x+2} + 8 = 4\sqrt{x+2} + 8$$

the left side and right side are the same so $x + 4\sqrt{x+2}$ is a valid solution
 For $(y-x)y' = y - x + 8$

implicit solution

an implicit solution does not solve for "y" alone instead it's mixed x and y together into one equation

ex:

$$x^2 + y^2 = 25$$

• this has x and y in the same equation

explicit solution

an explicit solution gives "y" directly as a function of x it has the form:

ex:

$$y = f(x)$$

$$y = x + 6\sqrt{x+3}$$

• notice this only has "x" in the function

Note:

a solution of a D.E that is free of parameters is called a particular solution
 a general solution has parameters example of general solution:

$$y = c_1 e^x + c_2 e^{2x}$$

$\underbrace{\hspace{1.5cm}}$
 parameters

Complementary solution

" y_c " this has a zero on it right side it also has a natural response (in simpler words the system is acting on its own)

ex: $y'' + 2y' + y = 0$

it has a zero so
there is no external
force its acting on
its own

Particular solution

" y_p " it does not have a zero on the right also it has a forced response something is driving it example of this force is a push, vibration, engine, etc

ex: $y'' + 2y' + y = e^x$

this is the driving
term

when you add the complementary and particular solution you get the physical motion of the system in simple term when you add y_c and y_p you get what you actually observe in real life

$$y = y_c + y_p$$

- y_c = how the system behaves on its own (natural motion, usually decays)
- y_p = how the system behaves because it is being driven
- y = what you actually observe in real life

homogeneous

the equation is homogeneous when it's equal to zero. So a complementary solution is homogeneous.

non-homogeneous

the equation is non-homogeneous when it's not equal to zero. So a particular solution is non-homogeneous.

examples:

Find the values of m so that the function $y = e^{mx}$ is a solution of $y'' - 5y' + 6y = 0$.

$$\left. \begin{array}{l} y' = me^{mx} \\ y'' = m^2 e^{mx} \end{array} \right\} \text{• Plug these into } y'' - 5y' + 6y = 0$$

$$(m^2 e^{mx}) - 5(me^{mx}) + 6(e^{mx}) = 0$$

You can factor e^{mx} from all the terms

$$e^{mx} (m^2 - 5m + 6) = 0$$

Set both the two equal to zero and solve for m .

$$e^{mx} = 0$$

↓
cannot solve

$$(m^2 - 5m + 6) = 0$$

$$(m-3)(m-2)$$

$m=3$	$m=2$
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$$y = e^{3x}, \quad y = e^{2x}$$

$$y = x^m, \quad x^2 y'' - 7x y' + 15y = 0$$

$$\left. \begin{aligned} y' &= m x^{m-1} \\ y'' &= m(m-1) x^{m-2} \end{aligned} \right\} \begin{array}{l} \bullet \text{ Plug this into the equation} \\ \text{and Solve For } m \end{array}$$

$$x^2 (m(m-1) x^{m-2}) - 7x (m x^{m-1}) + 15 (x^m) = 0$$

all of these terms have x^m in common but there is a problem two out of the three terms have other stuff attached to it but there is a solution $x^{m-2} \Rightarrow x^m \cdot x^{-2}$ because $x^{a-b} \Rightarrow x^a \cdot x^{-b}$ you can add the power if they have the same base

$$\underbrace{x^2 x^m x^{-2}}_{x^0} (m^2 - m) - 7 \underbrace{x x^m x^{-1}}_{x^0} (m) + x^m (15)$$

$$x^m (m^2 - m - 7m + 15)$$

$$x^m (m^2 - 8m + 15)$$

$$\begin{array}{l} x^m = 0 \\ \downarrow \\ \text{cannot solve} \end{array} \quad \begin{array}{l} (m^2 - 8m + 15) = 0 \\ (m-3)(m-5) \\ \boxed{m=3 \quad m=5} \end{array}$$

$$y = x^3, \quad y = x^5$$

Find values of m so that the function $y = e^{mx}$ is a solution of the given differential equation

$$y' + 2y = 0$$

$$y' = me^{mx}$$

$$(me^{mx}) + 2(e^{mx}) = 0$$

• factor e^{mx} out the function and solve for m

$$e^{mx} (m + 2) = 0$$

$$e^{mx} = 0$$

$$m + 2 = 0$$

↓
Cannot be
solve

$$\boxed{m = -2}$$

$$y = e^{-2x}$$

Verify that the indicated function is explicit solution of the given DE

$$6y' + y = 0, \quad y = e^{-x/6}$$

$$y' = -\frac{1}{6}e^{-x/6}$$

• Plug into the equation to verify

$$\cancel{6} \left(-\frac{1}{\cancel{6}} e^{-x/6} \right) + (e^{-x/6}) = 0$$

$$\cancel{-e^{-x/6}} + \cancel{e^{-x/6}} = 0$$
$$0 = 0$$

So, yes this equation is a solution for the DE

Verify that the function is a explicit solution of the DE

$$\underbrace{(y-x)y'}_{\text{left}} = \underbrace{y-x+18}_{\text{right}}, \quad y = x + 6\sqrt{x+3}$$

$$y' = 1 + 6(x+3)^{-1/2} \Rightarrow y' = 1 + \frac{6}{2}(x+3)^{-1/2}$$

$$y' = 1 + \frac{3}{\sqrt{x+3}}$$

• Plug into the equation to verify the solution

$$(y-x)y' \Rightarrow (\cancel{x} + 6\sqrt{x+3} - \cancel{x}) \left(1 + \frac{3}{\sqrt{x+3}}\right)$$

$$(y-x)y' \Rightarrow (6\sqrt{x+3}) \left(1 + \frac{3}{\sqrt{x+3}}\right)$$

$$(y-x)y' \Rightarrow 6\sqrt{x+3} + \frac{18\cancel{\sqrt{x+3}}}{\cancel{\sqrt{x+3}}}$$

$$(y-x)y' \Rightarrow 6\sqrt{x+3} + 18$$

left side

$$y-x+18 \Rightarrow (\cancel{x} + 6\sqrt{x+3} - \cancel{x}) + 18$$

$$y-x+18 \Rightarrow 6\sqrt{x+3} + 18$$

right side

left = right

$$6\sqrt{x+3} + 18 = 6\sqrt{x+3} + 18$$

the left side and right side are equal so the solution is valid for the DE

add-on notes

linear any non-linear have nothing to do with graphing linear and non-linear
Just explain to you how simple or how complex an differential equation will
be

Think of it like this:

- **Linear** → the DE is "friendly" in the sense that you can add two solutions together (superposition works).
- **Nonlinear** → the DE is "messy" because solutions interact with each other, and superposition doesn't work.